2. Python SQLite Game Shop POS

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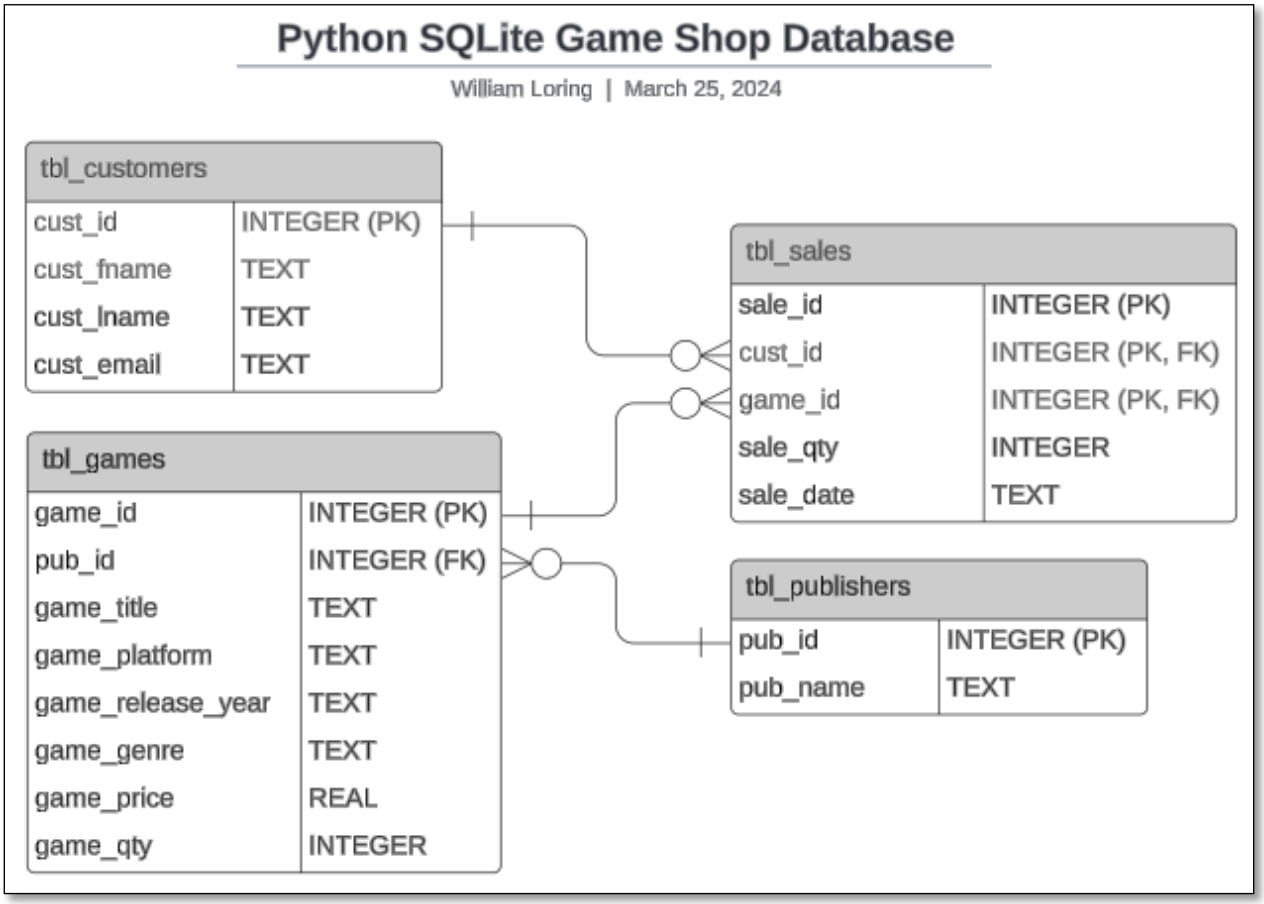
Time required: 60 minutes

- Follow all directions carefully and accurately.
- Think of the directions as minimum requirements.



Based on our new ERD, we need to add a sales table.

db_sales.py

```python
"""
    Name: db_sales.py
    Author: William Loring
    Created: 10/07/24
    CRUD module for game sales transactions
"""
# Import sqlite3 database library
import sqlite3

DATABASE = "game_shop.db"

# --------------------- SQL STATEMENTS ------------------------------------ #
# SQL statements are text. SQL queries can be very long.
# You can create a SQL statement and assign it to a string variable.

# ------------------------- CREATE TABLE ---------------------------------- #
# Create a table to store sales transactions
# Each sale record has a unique identifier, customer ID, game ID,
# quantity sold, and sale date
CREATE_TABLE = """
    CREATE TABLE IF NOT EXISTS tbl_sales (
    sale_id INTEGER PRIMARY KEY,  -- Unique identifier for each game
    cust_id INTEGER,              -- Foreign key referencing customers table
    game_id INTEGER,              -- Foreign key referencing games table
    sale_qty INTEGER              -- Number of copies sold
    sale_date INTEGER,            -- Date of sale
    FOREIGN KEY (cust_id) REFERENCES tbl_customers(cust_id),
    FOREIGN KEY (game_id) REFERENCES tbl_games(game_id)
    CONSTRAINT fk_customer
        FOREIGN KEY (cust_id)
        REFERENCES tbl_customers(cust_id)
        ON DELETE CASCADE
    CONSTRAINT fk_game
        FOREIGN KEY (game_id)
        REFERENCES tbl_games(game_id)
        ON DELETE CASCADE
);
"""
```

Main App Modification

Make the following changes to your main application by adding the reference to the new **db_sales** module and **db_sales.create_table()** method.

```python
"""
    Name: game_shop_pos_1.py
    Author: William Loring
    Created: 02/16/25
    Use SQLite with Python
"""

import tabulate
import db_app
import db_games
import db_publishers
import db_sales

MENU_PROMPT = """------  Game Shop Point of Sale  ------
(1) Add publisher
(2) Display all publishers
(3) Add new game
(4) Display all games
(5) Display all games and publishers
(6) Delete publisher
(7) Delete game
(9) Exit
Your selection: """


def main():
    db_publishers.create_table()
    db_games.create_table()
    db_sales.create_table()
    menu()
```

Menu Re Work

We are going to have a lot more choices in this new database. Time to redesign our menu system.

```python
MENU_PROMPT = """------  Game Shop Point of Sale  ------
(1) Table Maintenance
(2) Sales Transactions
(3) Reports
(9) Exit
Your selection: """

TABLE_MAINTENANCE_PROMPT = """------  Table Maintenance  ------
(1) Insert Publisher
(2) Insert Game
(3) Delete Publisher
(4) Delete Game
(9) Exit
"""

SALES_TRANSACTIONS_PROMPT = """------  Sales Transactions  ------
(1) Add Sale
(2) Display Sales Transactions
(3) Delete Sales Transaction
(9) Exit
"""

REPORTS_PROMPT = """------  Reports  -------
(1) Sales by Publisher
(2) Sales by Game
(3) Sales by Genre
(4) Sales by Platform
(5) Display Publishers
(6) Display Games
(7) Display Publishers & Games
(8) Display Sales Transactions
(9) Exit
"""
```

Move your existing menu items into the new menu structure using the nested pattern shown.

```python
def menu():
    while True:
        user_input = input(MENU_PROMPT)

    # ---------------------- TABLE MAINTENANCE ---------------------------- #
        if user_input == "1":

            # ---------------------- INSERT PUBLISHER ---------------------- #
            user_input = input(TABLE_MAINTENANCE_PROMPT)
            if user_input == "1":
                display_all_publishers()
                # Get publisher data from user
                pub_name = input("Enter publisher name: ")
                pub_country = input("Enter publisher country: ")

                # Add publisher to database
                db_publishers.add_record(
                    pub_name,
                    pub_country
                )

                display_all_publishers()
```

Assignment Submission

1. Attach the program files.
2. Attach screenshots showing the successful operation of the program.
3. Submit in Blackboard.